Business Sensory Platform

Delivering Business Agility by Optimizing Corporate DNA

Key Business Stakeholders

- Strategic
  - Who: Executive, marketing, product development
  - What: Identify opportunities, own risk, provide direction, approve investment
- Operations
  - Who: Core Functions, Supporting Functions (per business model)
  - What: Deliver core services to provide products and/or services to customers / consumers
  - What: Deliver supporting services to core service functions & external stakeholders
- Execution
  - Who: Anyone / anything involved in change (direct or indirect)
  - What: Planning, Designing, Managing, Funding, Approving, Building, Testing, & Implementing

Senses

- Short Term Sensors
  - Market changes, Competitive Threats, Compliance Updates, Supplier / Partner Changes

- Long Term Sensors
  - Market trends, Political Changes, Environment Changes, Behavioral changes, Fads
  - New Technologies
  - New Materials

Problem Classifications

- Quality
- Performance
  - Productivity
  - Financial
- Resource Optimization / Utilization
- Opportunity & Investment Planning (1)
- Risk Management (2)

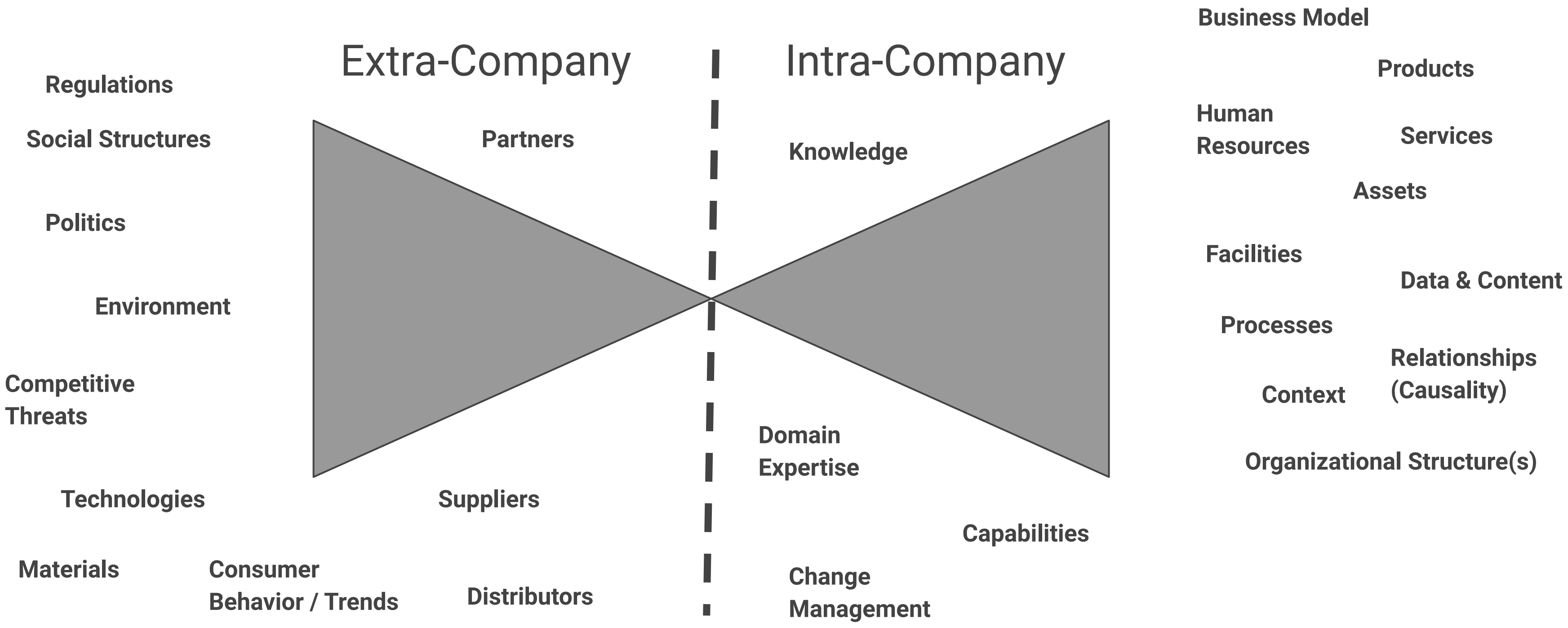
(1) Specific to Strategic Stakeholders
(2) Universal for all stakeholders

All other problems originate in the tactical / operational levels and aggregate (horizontally & vertically)

Sensory Model

Start with the Basics

- Sense - what is going on (intra- company) and/or (extra-company)
- Analyze - what do one or more sets of sensory data mean (understanding)
  - In some cases we can map these to known patterns
  - In some cases we may need additional data to identify an unknown pattern
  - In some cases, there may not be a pattern so we have to infer one or more patterns
- Respond - what action(s) should be taken once we understand
  - Decision(s) need to be made, which may create a forward / backward chaining effect
  - Each Decision needs to be mapped to impacts as part of scenario modeling / learning
- Learn - explore & improve decisions, impacts from decisions, & patterns



Extra-Company
Intra-Company
Regulations
Social Structures
Politics
Environment
Competitive Threats
Technologies
Materials
Consumer Behavior / Trends
Partners
Suppliers
Distributors
Knowledge
Domain Expertise
Capabilities
Change Management
Business Model
Products
Human Resources
Services
Assets
Facilities
Data & Content
Processes
Relationships (Causality)
Context
Organizational Structure(s)

Domain - Definition

- A domain is a conceptual boundary for a set of functional capabilities. Domains are useful structures for clarifying logical boundaries for:
    - Sensing change (intra & inter domains)
    - Measuring (performance, assessing impact, informing decisions)
    - Decision Making (intra domain changes that impact operations)
        - Episodically, temporally, and perhaps permanently.
        - Note: These events can impact "other" domains (vertical, horizontal)
    - Operational Execution (functions specific to the domain)
    - Infrastructure Services (enabling Capabilities & Functions required to support the Domain Specific Services (prior bullet points).

Domain Services Model

Capability (aligned with Domain)

- Activities (Events - Discrete, Processes - Flows, Cyclical Processes - Lifecycle)
- Decisions (only within the context of the domain)
- Services (specific to the domain : we’re not excluding shared services)
- Data (Explicit & Tacit) (operational, transactional, content, info, knowledge)
- Metadata (Common Dimensions + Context = stakeholder perspectives)
- Resources (human, machine = logical & physical)
- Models (incorporates the data & context into problem frameworks)
- Analytics (applies one or more sets of analyses to models)

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Domain Services

Sensors

Analytics Structures & Models

Foundation Services

Infrastructure Services

Infrastructure Services

Biz Arch Model (Cartography)

Adapted To Business Context & Capability Instances

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Biz Arch Model (Cartography)

Biz Arch is realized using structured set of models to build a cartographic map for:

- Capability Map
- Strategy Map
- Operational Map(s)
- Organization Map(s)
- Lifecycle Map(s)
- Biz Model Map
- Biz Canvas Map
- Biz Scenario Map(s)
- Heat Map

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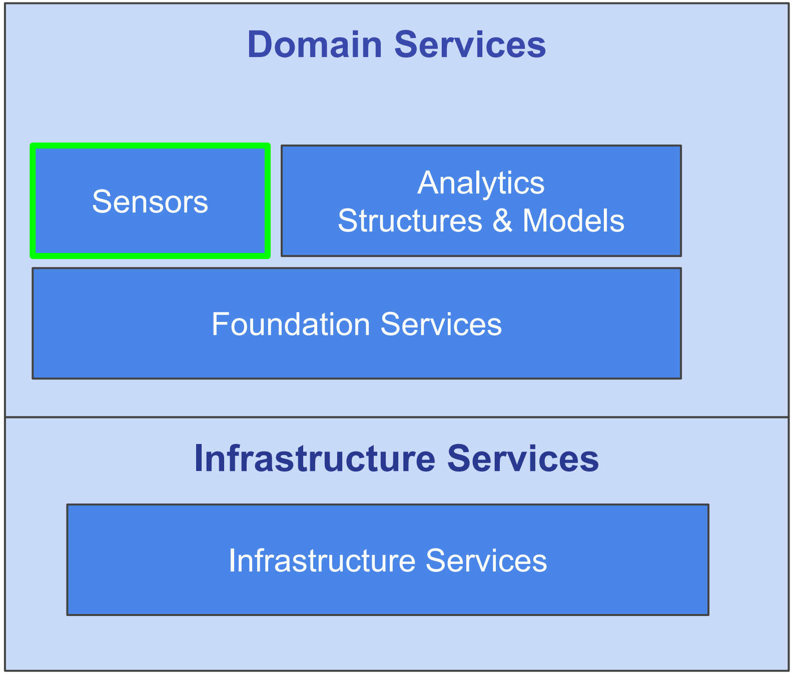

Sensors Include:

- Intra-Domain Sensors
  - Changes in data, services, triggers, events, behaviors, processes (ordered flow of events), analytics, outputs, and gaps with outcomes (from stakeholders).
  - Changes can be planned or unplanned.
  - Sensors are tuned for short-term changes and long-term changes (slowly evolving).
- Inter-Domain Sensors
  - Same as above only focused on changes from other Domains (horizontal, vertical).

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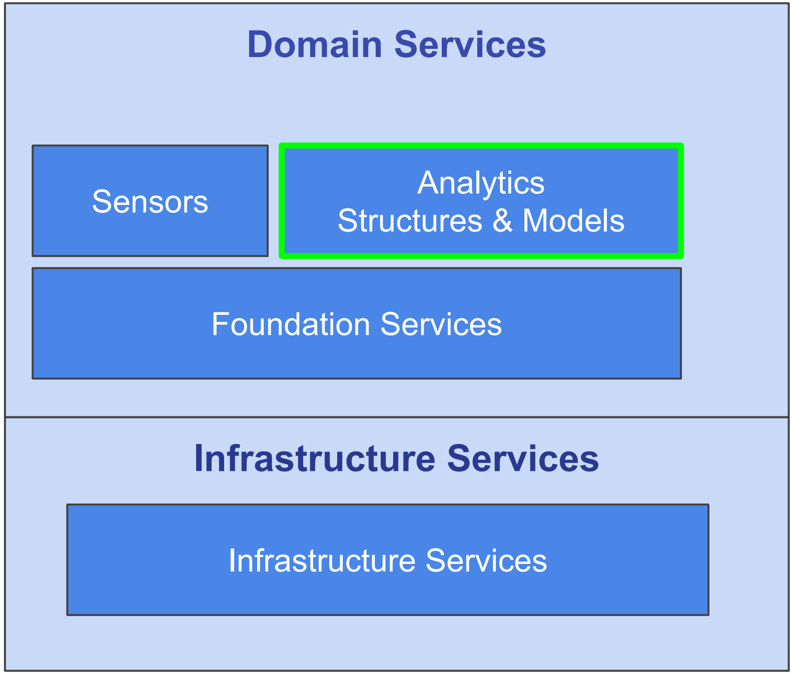

Analytics, Structures, & Models Include:

- Modeling Platform includes:
  - Catalog contains the specific terms and relationships drawn from the larger Biz Arch model (Graph DB).
  - Models & Analytics (including wizards) are filtered and tuned to the inputs from the domain sensors (intra & inter).
  - ML & AI run in discover mode, identifying patterns of causality from changes within and between domain (from sensors).
  - Event triggers

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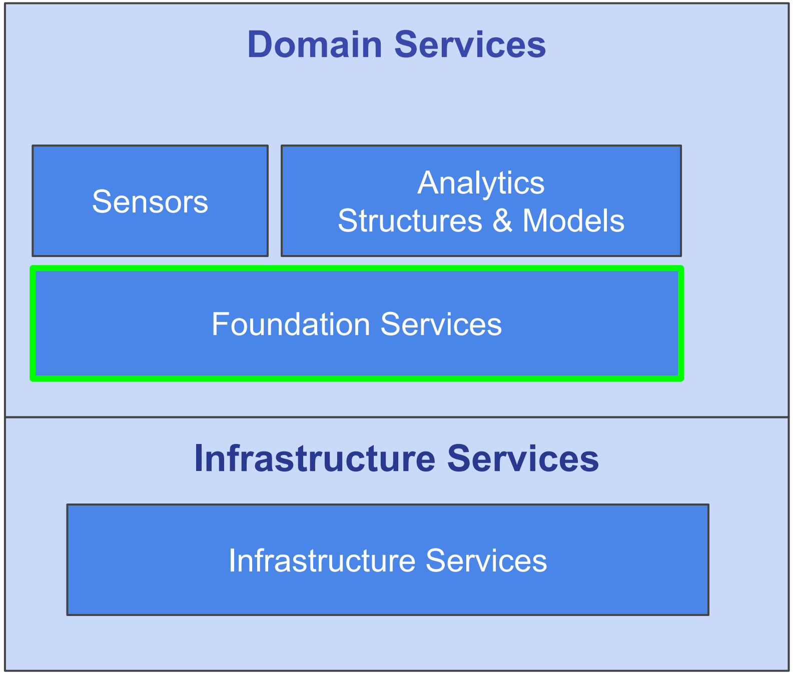

Foundational Services Includes:

- Pub / Sub
- Visualization
- Event / Workflow Management
- Decision Management
- User Experience & Business Scenario
- Analytics & Measures

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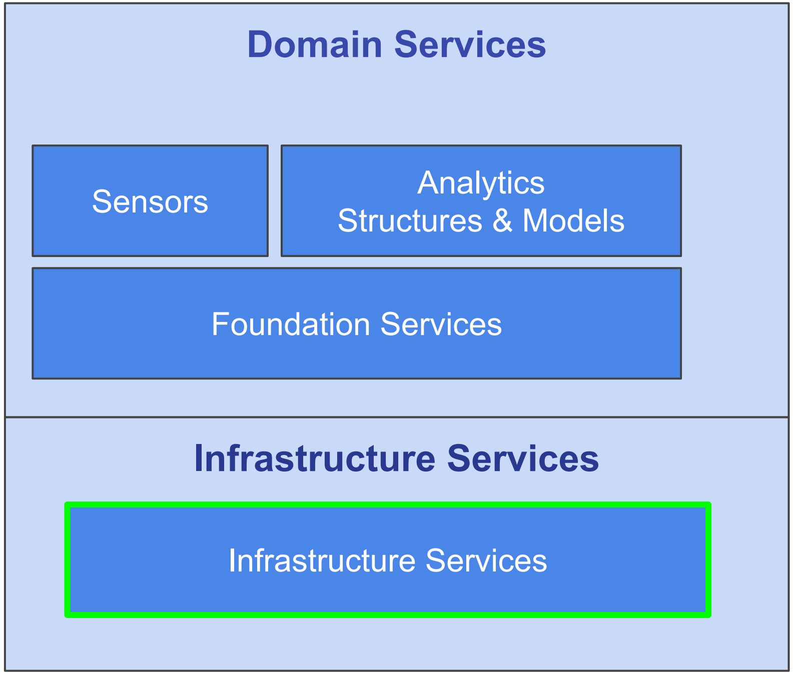
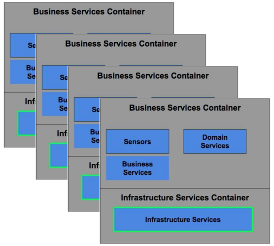


Infrastructure Services Includes:

- IoT Deployment
- Edge Based Computing
- Data Management Platform
  - Integration
  - Data Integrity
- MicroServices
- Registry Services
- AI & ML
- Cryptography & Blockchain
- Security, Resiliency, & Insulation

Deployment Model (domain specific)

Goal is to design the platform using a component design to support rapid integration, deployment, and tailoring for EACH domain (capability) using a progressive model.

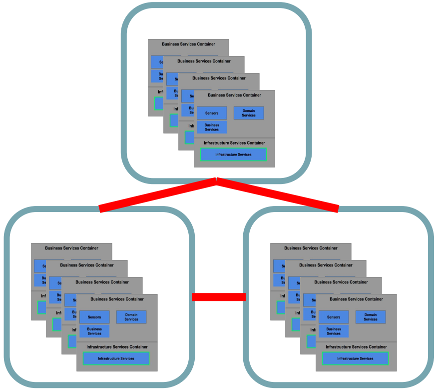


- A set of containers (business & infrastructure services) would be deployed for each domain.
- The sensory model within each domain would be tailored within the boundaries and aligned with the business context for the company (business unit ... TBD).

Deployment Model (Inter-Domain)

The key challenge is to identify where there are dependencies (relationships) via impacts between domains. These “edges” may be horizontal or vertical and each will need to be addressed as we deploy horizontal and/or vertical sensory capabilities:

- Need to determine how best to deploy these “extra” containers to warehouse the necessary (business & infrastructure services).
- Need to determine how to tailor the sensory model across domains while preserving the business context for the company (business unit ... TBD).



Implementation Planning

Each Step will be articulated in one or more slides in this section. Once complete, we can discuss tools / tech that we may want to leverage to reduce cost & time for deployment & tailoring

1. Define Capability Map (1-3 Levels)
2. Outline steps to map Generic Capability Map to Business Context
3. Identify 3 scenarios (see problem classification) for different domain levels
    a. Quality, Performance, Opportunity
4. Model each scenario
    a. Relationships & Attributes
    b. Outlining impact & context
5. Provide stepwise walkthrough for developing causal factors (by domain)
    a. Within each Domain
    b. Across Domains (Horizontal)
    c. Across Domains (Vertical)
6. Outline Sensory Model
    a. Data sensing, Analysis, Activities
    b. Options / Decision & Impacts

Define Capability Map (1-3 Levels)

Level 1	Level 2	Level 3	Level 4
Commercial Management	Marketing	Marketing Strategy	Plan/Develop Marketing Strategy
Commercial Management	Marketing	Marketing Strategy	Agency Performance/management
Commercial Management	Marketing	Market Research and Competitive Intelligence	Market research
Commercial Management	Marketing	Market Research and Competitive Intelligence	Competitive intel

Map Generic Capability Map to Business Context

Business Context can be captured in a properly built metadata model. The model can be derived from a variety of sources and incorporated into a structure. Some of the sources are external, most however can be found internally:

- Structured content - processes, systems, data, policies
- Unstructured content - documents, social media, email, communications

The challenge is to incorporate the context (variances in the use of terms and meaning from structured & unstructured sources) into a Metadata structure to understand the relationships.

Map Generic Capability Map to Business Context

There are some available tools for mining structured & unstructured sources and feeding these into a context and/or semantic engine to create a structure. That structure needs to be built and maintained as part of an organic (fluid) process.

Once in place, the next step is to align the reference model for a generic capability map to the Business Context Map (the company's metamodel) and maintain these mappings using (ideally) some form of ML (not rules as these are rigid and need lots of manual effort). Key Differentiator

This mapping between structures represents a virtual digital twin (of sorts), which can be used for evaluating scenarios and incorporating relational dependencies.

Tools & Technologies

- Supply Chain Sensors (Iot) (Leverage Azure / AWS cloud)
- Decision Management Engine (A/S - I will discuss)
- Metadata Map (Semantic Engine - I will discuss)
- Visualization (leverage Visualization Tools and/or A/S)
- Capability Map (Graph DB - Neo4j)
- Dependencies (Impacts) (ML Feed into Graph DB)
- Scenario Modeling (Decision thru Impact Lineage)
- Extending Capability Map to Business Context (Metadata Map)

3 Scenarios

- Quality
  - Domain - Commercial Management | Marketing | Market Research and Competitive Intelligence
  - Scenario - Market research (Invest $500K in customer focus study)
- Performance
  - Domain - Commercial Management | Marketing | Market Research and Competitive Intelligence
  - Scenario - Market research (Predictive Analytics for consumer insights is consuming increasing amounts of resources)
- Opportunity
  - Domain - Product Management | Product Development
  - Scenario - Purchase startup (evaluate M&A - buy decision)

Scenario Modeling

Scenario - Market research (Invest $500K in customer focus study)

Data Structures & Attributes	Business Capabilities	Analytics	Models
Market Segments - Industry - Demographic - Ethnographic	Product / Service Strategy Value Proposition Business Unit	4 Box risk factors 10 box impact vs likelihood Correlation matrix (risk) Correlation matrix (features)	Risk Assessment Strategy to Risk Risk to Product Product to Data
Customer - Desired Outcomes - Problems	Information Context Capability Outcome	Correlation Matrix (outcomes) Inference Matrix(s) (impact)	Product to Strategy Product Design to Features Features to Desired Outcomes
Product / Service - Ideas - Features	Stakeholder - Customer - Supplier	Portfolio Optimization - Product - Projects	Product to Services Services to Desired Outcomes
Issue Management Location			Desired Outcomes to Jobs Market Segments to Outcomes

Understanding Impact

Provide stepwise walkthrough for developing causal factors (by domain):

Within Core Domain

- Marketing Management

Across Domains (Horizontal)

- Product Management
- Customer Management
- Commercial Management
- Financial Management

Across Domains (Vertical)

- Strategic Management
- Product Development

Sensory Model

Data sensing, Analysis, Activities
Options / Decision & Impacts

The key is to identify the “type” of problem you’re dealing with using a recursive approach for applying a combination of sensors, analysis, decision modeling.

Problem Classification

High Complexity
- Sense
- Probe
- Analyze
- Decide
- Respond

Low Complexity
- Sense
- Analyze
- Decide
- Respond

Infinite Complexity
- Respond (Slowly)
- Probe
- Assess
- Analyze
- Decide
- Respond

No Complexity
- Sense
- Decide (Rules)
- Respond

Analytic Engine

Consists of 3 Components:

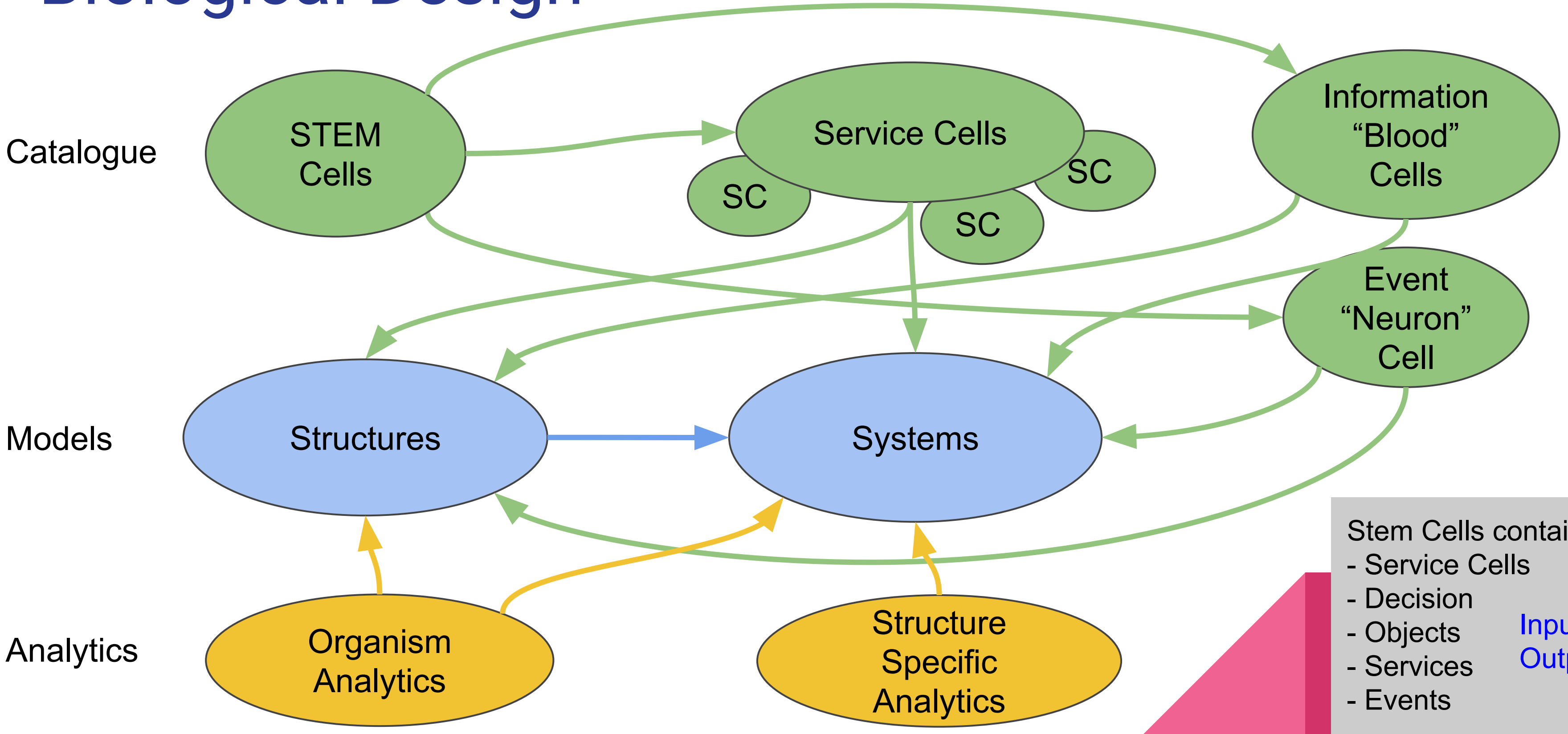
- Catalogue - taxonomies (domain specific)
- Models - systems & structures (domain specific)
  - Pairs of things (see. Anatol Holt)
- Analytics - subset of master set
  - Lists (ordered)
  - Flows (ordered list with beginning & end)
  - Nets (assemblage of flows - supports forward & backward chaining)
  - Trees (hierarchical matrices)
  - Base (matrices - base)

Any of the 3 components may have :

(1) Attributes
(2) Enumerations (values)

Biological Design

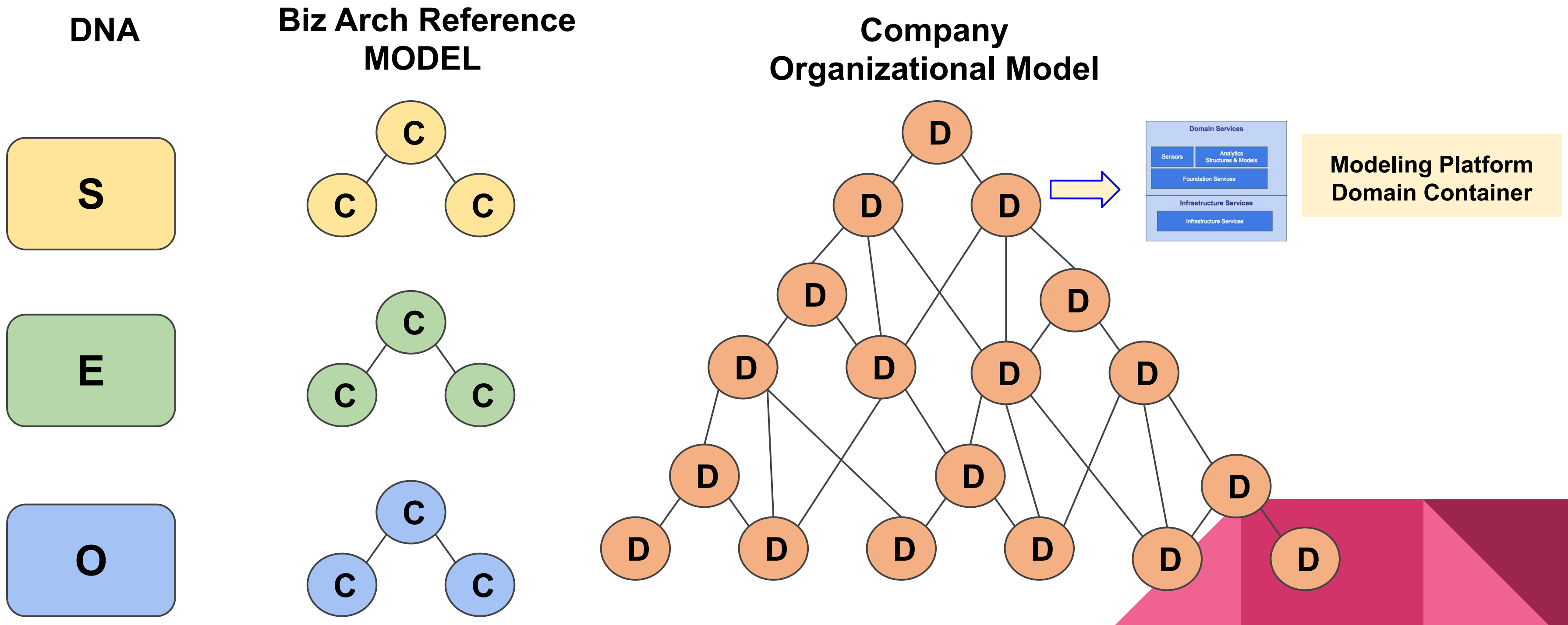
Like any biological entity, the system "learns" and adapts its behaviors by making decisions, taking action and responding to patterns of impact (desired & undesired outcomes)



Stem Cells contain:
- Service Cells
- Decision
- Objects
- Services
- Events

Inputs
Outputs

Organizational DNA Mapping



Paradigm Shift in Design & Decision Making

1980 - 2000	2000 +
• Design 80% Locked • Relationships Few (designed into RDB) • Data (Explicit) • Structures & Rules well understood • 20% Uncertainty • Design Prescriptive • Decision Making Centralized • Control, Risk, Impact, & Governance controlled manually intensive meetings (via horizontal & vertical Org. alignment across departments)	• Design 20% Locked • Relationships Many (ongoing discovery) • Data (Explicit & Tacit) • Structures Variable • Rules replaced by predictive analytics • 80% Uncertainty • Design Organic • Decision Making is decentralized • Control, Risk, Impact, & Governance controlled using augmented decision making (via horizontal & vertical Org. collaboration between domains)